

ATEF Z. ELSHERBENI
IEEE Fellow (2007) and ACES Fellow (2008)

Referred Journal Publications: (Total 243 Publications)

2026

- Fatih Kaburcuk and Atef Z. Elsherbeni, “Absorbed Power in Human Head Skin Due to Near-Field Exposure up to 100 GHz,” ACES Journal, vol. 41, no. 01, pp. 64-73, January 2026.
- Yunlong Mao, Deshun Li, Yuxuan Wang, Yifan Shen, and Atef Z. Elsherbeni, “A wideband filtering metasurface antenna with high edge roll-off rates based on characteristic mode analysis,” Journal of Electromagnetic Waves and Applications, vol. 40, Issue 6, 2026.
[DOI:10.1080/09205071.2025.2609162](https://doi.org/10.1080/09205071.2025.2609162)
- Kaikun Niu, Junqiang Wang, Yingsong Li, Zhixiang Huang, Xianliang Wu and Atef Z. Elsherbeni, “An Accurate Conformal Weakly Conditionally Stable FDTD Scheme and its Application on Electromagnetic Shielding Effectiveness Analysis,” IEEE Transactions on Antennas and Propagation, vol. 74, 2026. [DOI:10.1109/TAP.2026.3667697](https://doi.org/10.1109/TAP.2026.3667697)
- Matthew J. Dodd, Kobe Prior, Joseph E. Diener, Veysel Demir, and Atef Z. Elsherbeni, “A Full-Wave Method for Radar Cross-Section Analysis with Locally Generated Structured Electromagnetic Waves,” Applied Computational Electromagnetics Society (ACES) Journal, vol. 41, no. 3, pp. 207-215, March 2026.
- Ming Fang, Wei Yang, Yang Shi, Jian Feng, Ke Xu, Minglin Yang, Yingsong Li, Zhixiang Huang, and Atef Z. Elsherbeni, “3D-Topology-Coordinated Dynamic Load Balancing for Heterogeneous CPU + GPU Accelerated FDTD,” Microwave and Optical Technology Letters, 2026; <https://doi.org/10.1002/mop.70700>
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- Rob D. Jones, Jerome Cheron, Joseph E. Diener, Peter H. Aaen, Richard A. Chamberlin, Benjamin F. Jamroz, Dylan F. Williams, Ari D. Feldman, and Atef Z. Elsherbeni, “A Technique for Optimal On-Wafer Device Spacing at Millimeter-Wave Frequencies,” IEEE Microwave and Wireless Technology Letters, IEEE Microwave and Wireless Technology Letters, vol. 35, no. 3, pp. 370-373, March 2025.
- Donghua Zhou, Ming Fang, Jian Feng, Jie Zhao; Ke Xu, Yingsong Li, Zhixiang Huang, and Atef Z. Elsherbeni, “Spatial-Temporal Electromagnetic Field Simulation Based on Fourier Neural Operator,” IEEE Transactions on Antennas and Propagation, vol. 73, no. 9, September, 2025. [DOI: 10.1109/TAP.2025.3571529](https://doi.org/10.1109/TAP.2025.3571529)
- Guilin Hou; Guoda Xie, Yi Chen, Kaihong Song, Xuesong Deng, Ming Fang, Yingsong Li, Zhixiang Huang, and Atef Z. Elsherbeni, “A Unified Arbitrary-Order Symplectic FDTD(p,q) Algorithm Based on Matrix Exponential Method for Anisotropic Time-Varying Plasma and PML,” IEEE Transactions on Microwave Theory and Techniques, vol. 73, no. 8, pp. 4630-4646, August 2025.
- Yuxian Zhang, Yilin Kang, Naixing Feng, Xiaoli Feng, Zhixiang Huang, and Atef Z. Elsherbeni, “Scale-Compressed Technique in Finite-Difference Time-Domain Method for

Multi-Layered Anisotropic Media,” IEEE Journal on Multiscale and Multiphysics Computational Techniques, vol. 10, pp. 85-93, 2025.

- Yinfeng Xia, Yingsong Li, Wei Xue, Atef Elsherbeni, “A Passive Mixed Loaded Over-Hundred-Octave Miniaturized Super-Wideband Antenna,” IEEE Antennas and Wireless Propagation Letters, vol. 24, no. 8, August 2025. [DOI: 10.1109/LAWP.2025.3564970](https://doi.org/10.1109/LAWP.2025.3564970)
- Kaikun Niu, Wenlong Tao, Wenkai Zhang, Xingang Ren, Yingsong Li, Wentie Yang, Zhixiang Huang, Xianliang Wu and Atef Z. Elsherbeni, "Ultra-Low Frequency Electromagnetic Wave Analysis in Ocean-Ionosphere Based on ADI-FDTD Combined with SIBC," IEEE Transactions on Antennas and Propagation, 2025. [DOI: 10.1109/TAP.2025.3608798](https://doi.org/10.1109/TAP.2025.3608798)
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- Yi Chen, Guoda Xie, Guilin Hou, Ziheng Song, Yingsong Li, Zhixiang Huang, and Atef Z. Elsherbeni, “An 3D Unconditionally Stable CDI-FDTD Method based on the Conformal Technique for Simulating Complex Curved PEC Objects,” IEEE Transactions on Antennas and Propagation, 2025, Accepted
- Zhipeng Chen, Linqing Li, Chao Yang, Wei E. I. Sha, Yingsong Li, Zhixiang Huang, Xianliang Wu, Atef Z. Elsherbeni, and Kaikun Niu, “The 3D Unconditionally Stable Multisymplectic FDTD Method for the Computational Electromagnetic Simulation,” IEEE Transactions on Antennas and Propagation, 2025. [DOI: 10.1109/TAP.2025.3617119](https://doi.org/10.1109/TAP.2025.3617119)

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- Kaikun Niu, Wenkai Zhang, Minquan Li, Xingang Ren, Yingsong Li, Zhixiang Huang, Bo Wu, Xianliang Wu, and Atef Z. Elsherbeni, “Transient Electromagnetic–Thermal Co-Simulation of Microwave/RF Integrated Circuits by the HIE-FDTD Method,” IEEE Transactions on Microwave Theory and Techniques, vol. 72, no. 9, pp. 5228-5244, 2024. [DOI: 10.1109/TMTT.2024.3373767](https://doi.org/10.1109/TMTT.2024.3373767)
- Guocui Zhu; Kaikun Niu; Minquan Li; Yingsong Li; Xingang Ren; Guoda Xie; Zhixiang Huang; Xianliang Wu; and Atef Z. Elsherbeni, “High-Order ME-CFS-PML Implementations for Terminating the FDTD Domain Composed of Arbitrary Media,” IEEE Transactions on Microwave Theory and Techniques, vol. 72, no. 4, April 2024.
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- Collin T. Kringlen, Atef Z. Elsherbeni, and Jamal Rostami, “High Gain Circularly Polarized Patch Antenna for Communications Between Cutting Tools and the Control Unit of Mining Equipment,” ACES Journal, vol. 39, no. 7 pp. 642-650, 2024. [DOI:10.13052/2024.ACES.J.390708](https://doi.org/10.13052/2024.ACES.J.390708)
- Naixing Feng, Huan Wang, Yuxian Zhang, Zhixiang Huang, and Atef Z. Elsherbeni, “Alternative Implementation of EM Propagation for 3-D Layered Lossy Media by SMM Method,” IEEE Transactions on Antennas and Propagation, vol. 72, no. 8, pp. 6599-6613, August 2024. DOI: 10.1109/TAP.2024.3416053
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- Kaikun Niu; Zijing Xiao; Guoda Xie; Xingang Ren; Yingsong Li; Zhixiang Huang; Xianliang Wu; Atef Z. Elsherbeni, “A Stochastic FDTD Algorithm for Uncertainty Quantification of Electromagnetic-Thermal Simulation,” IEEE Transactions on Microwave Theory and Techniques, Early Access, 2023.
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- Alec Weiss, Atef Elsherbeni, Jeanne Quimby, and Jacob D. Rezac, “Monte Carlo Augmented Channel Estimator,” URSI Radio Science Letters, vol. 4, pp. 1-4, 2022. DOI: 10.46620/22-0006

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- Joshua M. Kast; Jacob D. Rezac; Stefania Römisch; Atef Z. Elsherbeni; Jeanne T. Quimby, “Timing Offset and Timing Stability for a Dual-Clock Channel Sounder,” IEEE Transactions on Instrumentation and Measurement, vol. 70, 2021. DOI 10.1109/TIM.2021.3082257
- Zhen Su; Kirill Klionovski; Hanguang Liao; Yiming Chen; Atef Z. Elsherbeni; Atif Shamim, “Antenna-on-Package Design: Achieving Near-Isotropic Radiation Pattern and Wide CP Coverage Simultaneously,” IEEE Transactions on Antennas and Propagation, vol. 69, no. 7, July 2021.
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- Andres Velasco, Atef Z. Elsherbeni, Joseph E. Diener, Mohammed F. Hadi, "Debye Parameters of Humidity-Varying Soils for Induction Logging Techniques," *Applied Computational Electromagnetics Society (ACES) Journal*, vol. 35, no. 9, pp. 979-984, 2020.
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